WEEK 1:

ARRAYS & BASIC HASHING

Key Topics:

- Prefix Sum
- Two Pointer Technique
- Sliding Window
- HashMap-based Problems

Must Solve Problems:

- Two Sum(LeetCode #1)
- Best Time to Buy and Sell Stock(LeetCode #121)
- Kadane's Algorithm(Maximum Subarray Sum) (LeetCode #53)
- Subarray Sum Equals K(LeetCode #560)
- Longest Consecutive Sequence(LeetCode #128)
- Product of Array Except Self(LeetCode #238)

WEEK 2:

SORTING & SEARCHING

Key Topics:

- Binary Search on Answer
- Custom Sorting
- Search Space Problems

Must Solve Problems:

- Binary Search (LeetCode #704)
- Find First and Last Position of Element in Sorted Array(LeetCode #34)
- Search in Rotated Sorted Array(LeetCode #33)
- Aggressive Cows / Allocate Books
- Kth Largest Element (LeetCode #215)
- Merge Sorted Arrays (LeetCode #88)
- Find Peak Element (LeetCode #162)

WEEK 3:

STRINGS

Key Topics:

- Sliding Window
- HashMap Counting
- Palindrome, Anagram, Substring

Must Solve Problems:

- Valid Anagram(LeetCode #242)
- Longest Substring Without Repeating Characters(LeetCode #3)
- Group Anagrams(LeetCode #49)
- Minimum Window Substring (LeetCode #76)
- Implement strStr() / KMP Algorithm(LeetCode #28)
- Count and Say(LeetCode #38)

WEEK 4:

LINKED LIST

Key Topics:

- Fast and Slow Pointers
- Reversal
- Cycle Detection
- Merging

Must Solve Problems:

- Reverse Linked List(LeetCode #206)
- Middle of the Linked List(LeetCode 876)
- Merge Two Sorted Lists(LeetCode #21)
- Remove Nth Node from End(LeetCode #19)
- Intersection of Two Linked Lists(LeetCode #160)
- Palindrome Linked List (LeetCode #234)

WEEK 5:

STACK & QUEUE

Key Topics:

- Monotonic Stack
- Next Greater Element
- Stack Simulation

Must Solve Problems:

- Valid Parantheses(LeetCode #20)
- Min Stack (LeetCode #155)
- Next Greater Element (LeetCode #496)
- Daily Temperatures (LeetCode #739)
- Implement Queue using Stack(LeetCode #232)
- Evaluate Reverse Polish Notation(LeetCode #150)
- Largest Rectangle in Histogram(LeetCode #84)

WEEK 6:

TREES (BINARY & BST)

Key Topics:

- Inorder, Preorder, Postorder
- Height, Diameter
- Lowest Common Ancestor
- BFS / DFS

Must Solve Problems:

- Maximum Depth of Binary Tree (LeetCode #104)
- Diameter of Binary Tree (LeetCode #543)
- Symmetirc Tree (LeetCode #101)
- Level Order Traversal (LeetCode #102)
- Lowest Common Ancestor(LCA) (LeetCode #236)
- Validate Binary Search Tree (LeetCode #98)
- Construct Binary Tree from Traversal (LeetCode #105)

WEEK 7 :

RECURSION & BACKTRACKING

Key Topics:

- Permutations
- Subsets
- Combination Sum

Must Solve Problems:

- Subsets (LeetCode #78)
- Combination Sum(LeetCode #39)
- Permutations (LeetCode #46)
- Sudoku Solver (LeetCode #37)
- N-Queens (LeetCode #51)
- Rat in Maze (N/A – common on GFG)
- Word Search (LeetCode #79)

WEEK 8:

DP & GRAPHS(INTRO)

DP Must Solve Problems:

- Climbing Stairs (LeetCode #70)
- House Robber (LeetCode #198)
- Coin Change (LeetCode #322)
- Longest Increasing Subsequence (LeetCode #300)
- 0/1 Knapsack (N/A – variant LeetCode #416)

Graph Must Solve Problems:

- BFS / DFS of Graph (N/A- basic traversal, similar to LeetCode #797)
- Number of Islands (LeeCode #200)
- Detect Cycle in Directed Graph (LeetCode #207)
- Topological Sort (LeetCode #210)
- Dijkstra’s Algorithm (optional advance)(LeetCode #743)